Lab 4: Incremental Pattern Mining & Correlation on YouTube Comments

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2. Objective

The objective of this lab is to simulate the incremental arrival of data by processing YouTube comments in segments (chunks). We aim to:

1. Mine frequent patterns (unigrams and co-occurring pairs) from each data chunk.
2. Track the evolution of specific word frequencies over time.
3. Analyze correlations between specific terms to understand thematic relationships in the comments.

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3. Tools and Libraries Used

- **Language:** Python 3.11
- **Libraries:**
  - pandas: For data manipulation and loading CSV files.
  - numpy: For splitting the dataset into chunks.
  - matplotlib.pyplot: For generating bar charts and trend plots.
  - itertools.combinations: For identifying co-occurring word pairs.
  - ast: For parsing string representations of lists.
  - collections.Counter: For counting word frequencies.

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4. Dataset Description

- **Source:** cleaned_comments.csv (derived from Lab 2).
- **Content:** YouTube comments with a preprocessed cleaned_tokens column containing lists of words.
- **Structure:** The dataset is treated as a stream of transactions and is segmented into **5 equal chunks** to simulate time-based data arrival.

5. Methodology / Procedure

5.1 Data Loading and Segmentation

The dataset was loaded using Pandas. The cleaned_tokens column was parsed using ast.literal_eval to ensure it was treated as a list of strings rather than a single string. The data was then split into 5 equal parts using numpy.array_split().

5.2 Pattern Mining per Chunk

For each of the 5 chunks, we iterated through the transactions to:

- Count the frequency of individual words (unigrams).
- Generate and count word pairs (bigrams) using itertools.combinations.
- The top 10 most frequent unigrams were visualized for each chunk.

5.3 Tracking Pattern Evolution

We selected specific interesting keywords (e.g., "zebra", "nature", "migration") and tracked their counts across all five chunks. This data was aggregated into a frequency table to visualize trends.

5.4 Correlation Analysis

Using the frequency data collected in step 5.3, we constructed a correlation matrix using pandas.corr(). This helps quantify how strongly the presence of one word relates to another across the different time chunks (e.g., does "migration" always appear when "zebra" appears?).

5.5 Reflection and Insight

We analyzed the resulting tables and plots to interpret why certain patterns peaked in specific chunks and how the words are semantically related.

6. Code Implementation

Python

```
import pandas as pd
```

```python
import numpy as np
import matplotlib.pyplot as plt
import ast
from collections import Counter
from itertools import combinations

# 1. Load Data
df = pd.read_csv('cleaned_comments.csv')
df['cleaned_tokens'] = df['cleaned_tokens'].apply(ast.literal_eval)

# 2. Split into 5 Chunks
chunks = np.array_split(df, 5)

# 3. Pattern Mining Function
def analyze_chunk(chunk_data, chunk_id):
    all_tokens = [token for sublist in chunk_data['cleaned_tokens'] for token in sublist]

    # Unigrams
    unigram_counts = Counter(all_tokens)

    # Plotting Top 10 Unigrams
    common_unigrams = unigram_counts.most_common(10)
    words, counts = zip(*common_unigrams)

    plt.figure(figsize=(10, 5))
    plt.bar(words, counts, color='skyblue')
    plt.title(f'Top 10 Unigrams in Chunk {chunk_id}')
    plt.xlabel('Words')
    plt.ylabel('Frequency')
    plt.show()

    return unigram_counts

# 4. Tracking & Correlation Analysis
keywords = ['zebra', 'beautiful', 'nature', 'migration', 'journey']
history = {word: [] for word in keywords}

for i, chunk in enumerate(chunks):
    counts = analyze_chunk(chunk, i+1)
    for word in keywords:
        history[word].append(counts.get(word, 0))

# Create DataFrame for trends
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trend_df = pd.DataFrame(history, index=[f'Chunk {i+1}' for i in range(5)])

# Correlation Matrix
correlation_matrix = trend_df.corr()
print(correlation_matrix)

# Plotting Trends
trend_df.plot(marker='o', figsize=(10, 6))
plt.title('Word Frequency Evolution Across Chunks')
plt.ylabel('Frequency')
plt.grid(True)
plt.show()
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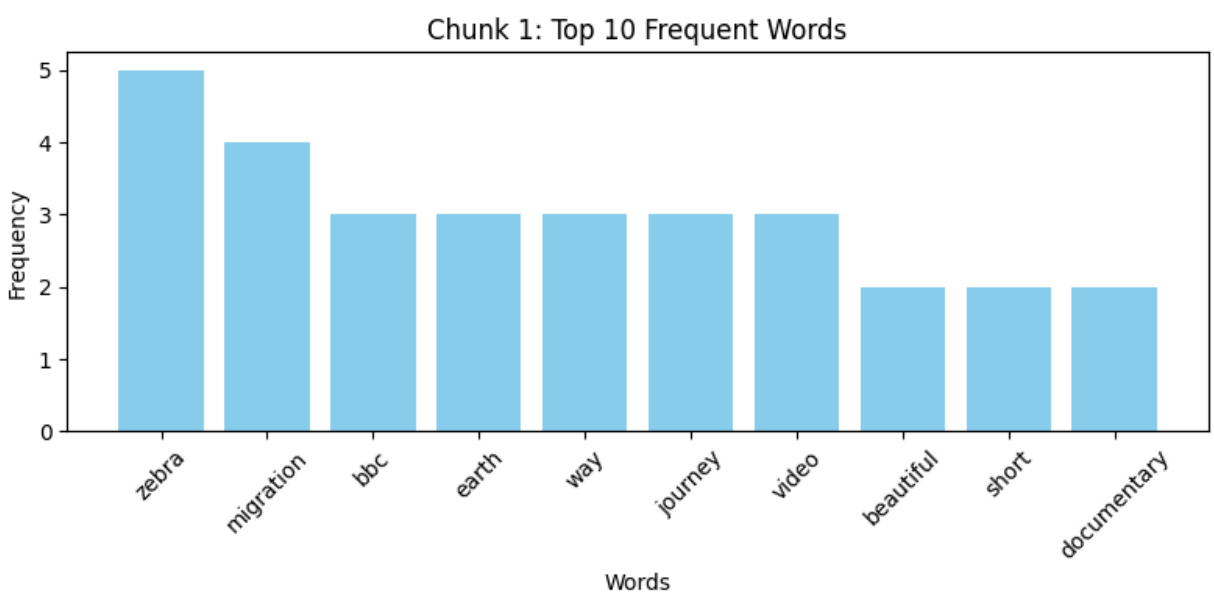
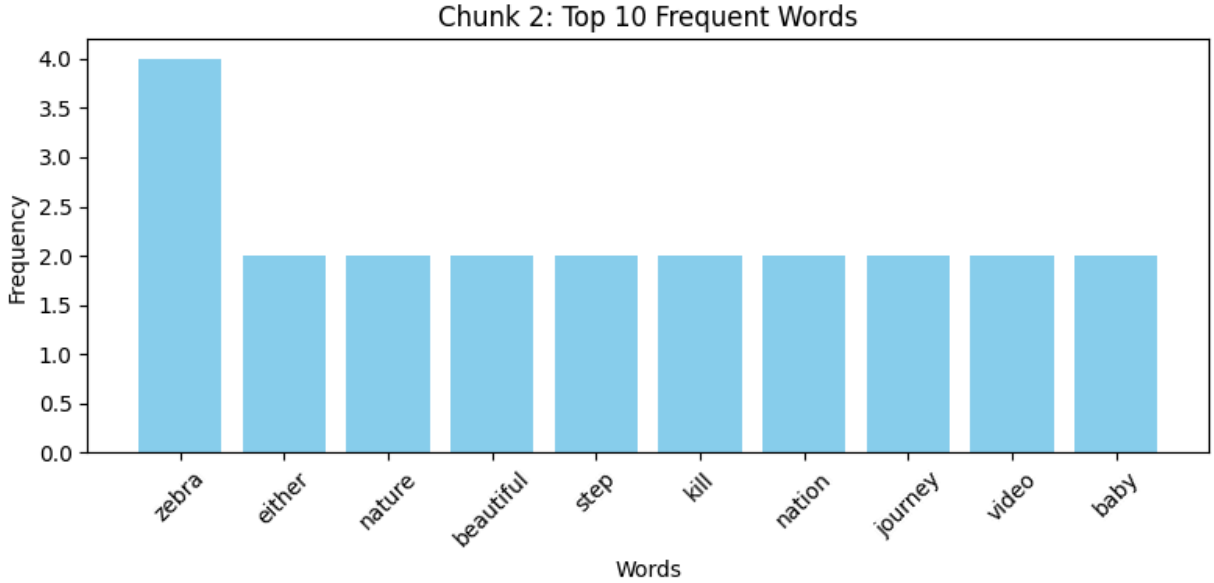
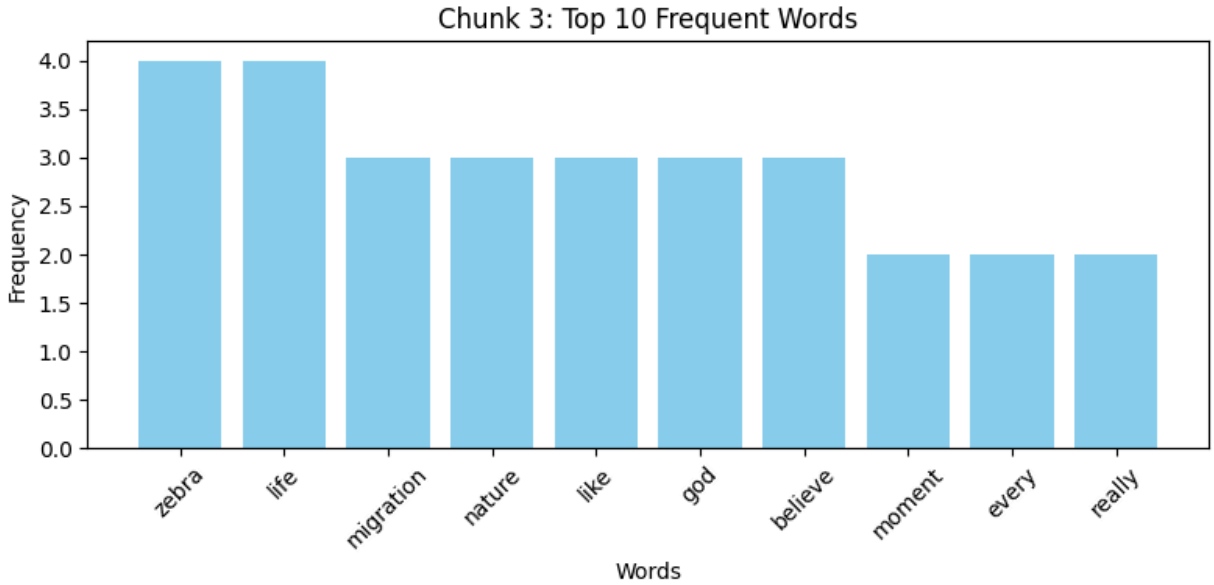
7. Results and Discussion

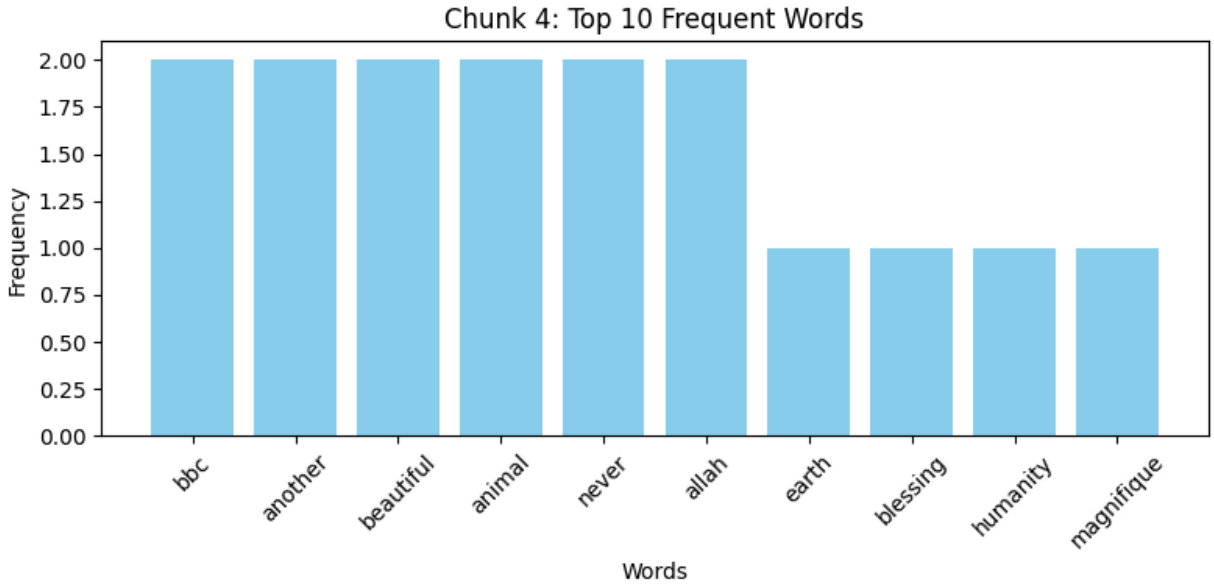
7.1 Unigram Distributions per Chunk

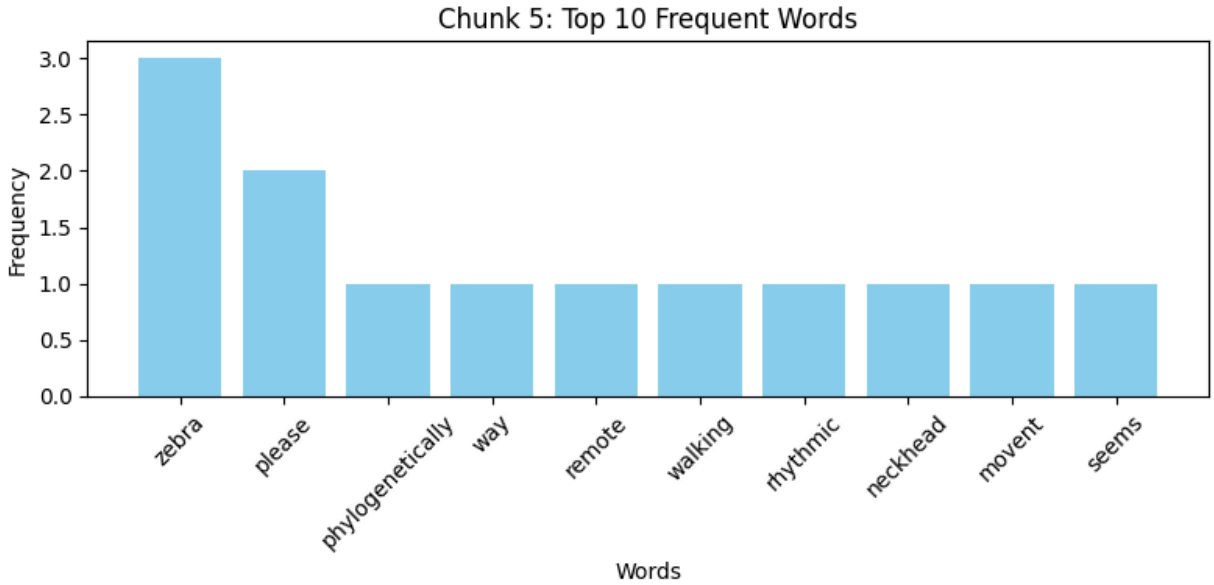
The following bar charts display the top 10 most frequent words found in each of the 5 data chunks.




Chunk 2: Top 10 Frequent Words
Frequency
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
zebra
either
nature
beautiful
step
kill
nation
journey
video
baby
Words


Chunk 3: Top 10 Frequent Words
Frequency
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
zebra
life
migration
nature
like
god
believe
moment
every
really
Words





7.2 Frequency Evolution

We tracked specific keywords ("zebra", "beautiful", "nature", "migration", "journey") to see how their popularity shifted over time.

Table 1: Keyword Frequency per Chunk

Chunk	zebra	beautiful	nature	migration	journey
Chunk 1	5	2	2	4	3

Chunk 2	4	2	2	1	2
Chunk 3	4	2	3	3	1
Chunk 4	1	2	1	0	0
Chunk 5	3	1	1	0	1

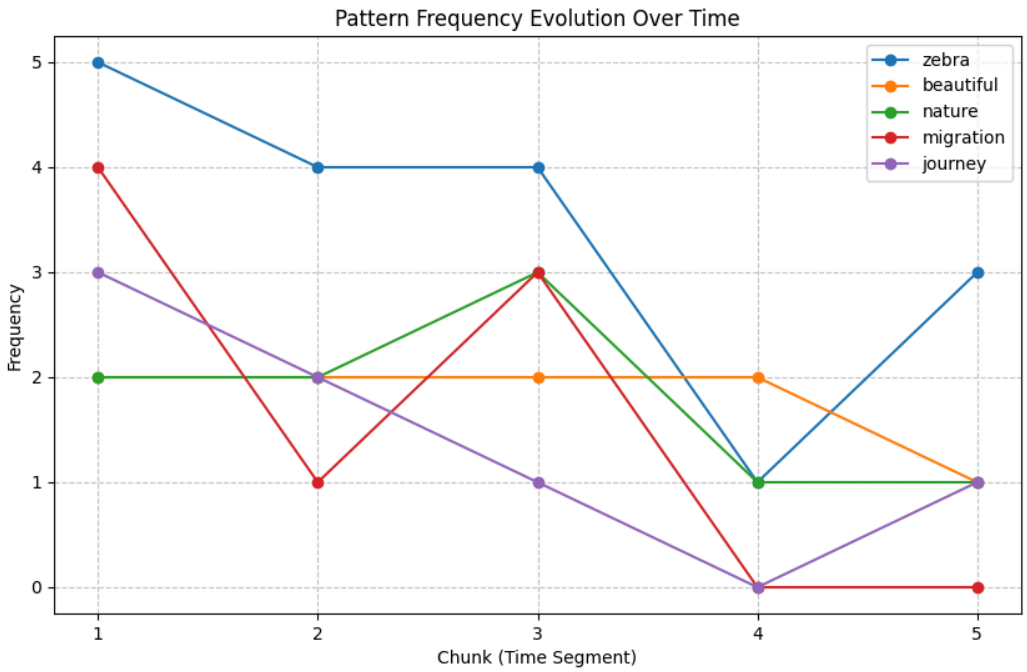


Figure 6: Frequency Evolution of Selected Keywords

7.3 Correlation Analysis

The correlation matrix below reveals how strongly the frequencies of these words are related.

Table 2: Correlation Matrix

	zebra	**beautiful**	**nature**	**migration**	**journey**
zebra	1.00	0.15	0.67	0.80	0.90
beautiful	0.15	1.00	0.53	0.49	0.20

nature	0.67	0.53	1.00	0.76	0.37
migration	0.80	0.49	0.76	1.00	0.70
journey	0.90	0.20	0.37	0.70	1.00

7.4 Discussion

- **Pattern Evolution:** The word "zebra" started with high frequency (5) in Chunk 1 but dropped significantly by Chunk 4 (1), suggesting the topic of discussion shifted away from specific animals. "Beautiful" remained relatively stable, indicating it is a general descriptive term used regardless of the specific topic.
- **Correlations:**
  - **High Correlation:** "Zebra" and "Journey" have a very high correlation (**0.90**). This suggests that when users discuss zebras, they are likely discussing their journey/migration context.
  - **Low Correlation:** "Beautiful" has a low correlation with "Zebra" (**0.15**). This implies that comments calling the video "beautiful" do not necessarily depend on the mention of "zebras" specifically; they might be praising the scenery or production quality instead.
- **Challenges:** One challenge was handling the sparsity of data in later chunks (e.g., Chunk 4 had 0 counts for migration and journey), which can skew correlation results.

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8. Conclusion

In this lab, we successfully simulated incremental pattern mining. We observed that while some general terms (like "beautiful") remain stable over time, specific topic words (like "zebra" and "migration") fluctuate together. The strong correlation between "zebra" and "migration" confirms that these terms are thematically linked in the user comments, likely referring to a specific event in the video content.